

Jaejin Cha

Software Engineer | ML Systems, Simulation & Planning

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Summary

Software engineer and M.S. Computer Science student with AI/data systems experience and research in autonomous systems. IROS 2026 author in multi-robot planning, with autonomous-driving simulation work in reinforcement and imitation learning.

Technical Skills

Languages: Python, C++, SQL; **Systems & Data:** Linux, Docker, Git, REST APIs, PostgreSQL, Apache Spark, HDFS, Oracle Database, OCI
ML & Planning: PyTorch, Gymnasium, reinforcement learning, imitation learning, Graphs of Convex Sets, multi-agent planning

Experience

Hewlett Packard Enterprise

Spring, TX

Software Engineer – AI Applications and Enterprise Systems

Intern: May 2025 – May 2026 | Full-time: Jun 2026 – Present

- Build production AI applications enabling operations teams to explore enterprise data through natural-language queries and generate on-demand insights.
- Implement text-to-SQL and tool-calling workflows using Oracle AI Database and PL/SQL packages, incorporating business-rule guardrails.
- Built a custom vector storage and similarity-search layer in Oracle 19c using PL/SQL, REST APIs, and embedding models.

Texas A&M University

College Station, TX

Graduate Researcher – Multi-Robot Planning & Coordination

Aug 2024 – Present

- Formulated cyclic multi-robot rendezvous as constrained shortest-path planning on a Graph of Convex Sets in phase-time space, enforcing timing and traversal-frequency feasibility across range-limited communication events.
- Built Python simulation and GCS planning infrastructure for multi-robot coordination; implemented graph sparsification to reduce planning complexity, evaluated against consensus baselines on an 11-robot topology, and validated trajectories on Robotarium hardware.

ENDEAVR Institute

College Station, TX

Machine Learning Engineer Intern

Oct 2024 – Feb 2025

- Implemented a PPO-based generator for a GAIL imitation-learning pipeline in PyTorch and validated it on Gymnasium environments.

University of Delaware

Newark, DE

Autonomous Driving Research Intern

Jun 2023 – Aug 2023

- Modified a Gymnasium-based highway simulator and implemented DQN and DDPG agents for discrete and continuous vehicle control.
- Ran reward-weight experiments on trade-offs between energy efficiency and driving metrics; presented results at a UD research symposium.

Texas A&M University

College Station, TX

Undergraduate Research Assistant

Sep 2022 – May 2023

- Configured Apache Spark and HDFS environments on Linux clusters and developed Python/Bash tooling to collect and analyze garbage-collection logs for runtime and memory profiling.
- Modified Peregrine source code to convert multi-label graph datasets into binary format for graph-processing workflows.

Publications

Jaejin Cha, Dylan Shell. “Planning and Execution for Multi-Robot Cyclic Rendezvous via Graphs of Convex Sets”. IROS 2026 (accepted).

Jaejin Cha. “Rendezvous-Based Multi-Robot Communication Under Cyclic Paths”. Master’s thesis (defended), 2026.

Selected Projects & Activities

AutonomyForge | C++20, Python

Aug 2026 – Present

- Designing and building a C++20 multi-agent simulation and planning framework for deterministic autonomy experiments, with modular interfaces for scenarios, planners, simulation, and evaluation.

Bitcoin Trading Simulator | Python, PyTorch, NumPy

Nov 2024 – Feb 2025

- Built an MDP-based Bitcoin trading environment from Kaggle data and trained reinforcement-learning agents for sequential decisions.

NatureNet | Python, Flask, PostgreSQL, YOLOv8

Jan 2024 – May 2024

- Built a YOLOv8 wildlife-detection app with Flask API inference, configurable alerts, and PostgreSQL-backed user settings and notifications.

Texas Aimbots | Computer Vision Developer

Sep 2023 – May 2024

- Implemented OpenCV-based target detection and contour extraction for RoboMaster University League competition robots.

Education

Texas A&M University

College Station, TX

M.S. Computer Science, GPA: 3.71/4.00 - Thesis in Multi-Robot Systems | Advisor: Dr. Dylan Shell

Expected Dec 2026

Texas A&M University

College Station, TX

B.S. Computer Science, Minor in Mathematics

Aug 2024